Low-Energy Electron Capture in Two-Dimensional Plasmonic Materials

A Dressed-Mass Mechanism, a Regime-Robust Rate Bound, and a Decisive Experimental Test

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Abstract

Converting a proton to a neutron by electron capture, $p + e^- \rightarrow n + \nu_e$, costs 0.782 MeV — roughly five orders of magnitude beyond any chemical, thermal, or condensed-matter energy scale. This single fact is why neutron-first proposals for low-energy nuclear reactions have remained contested. Strong-field quantum electrodynamics suggests one escape: an electron immersed in an intense coherent field acquires an effective rest mass $\tilde{m}c^2 = m_e c^2\sqrt{(1+\xi^2)}$, and the capture channel opens once the dimensionless field strength reaches $\xi \geq \xi_{th} \approx 2.32$. We develop this dressed-mass mechanism for a graphene/borophane geometry, in which terahertz acoustic plasmons confine the driving field most strongly, note that the Volkov mass is a plane-wave (null-field) result whose transfer to longitudinal plasmon near-fields is itself an assumption, and then ask — within assumptions stated explicitly — whether plasmon coherence can actually supply the 0.782 MeV deficit.

It cannot. The field must deliver the deficit in $N = \Delta E/\hbar\omega \approx 8\times10^7$ quanta, and the natural rescue — treating the plasmon as a coherent N-quantum object, whose required occupation appears to fall toward unity — fails for two independent reasons. First, the N-quantum coupling is not free: generated at N-th order it scales as $g_{eff} \sim g_0^N$, which exactly cancels the n^N coherence bonus and restores the multiphoton suppression $|J_N(\xi)|^2$. Second, the required occupation cannot fall below N at all, because a^N annihilates N quanta. The capture rate is therefore bounded tens of thousands of orders of magnitude below detectability even in the most favorable accessible case — and the bound is regime-robust: a Keldysh analysis shows that the quasistatic (tunneling) channel, the least suppressed available, still falls short by $\sim6\times10^4$ orders, with every multiphoton channel failing by far more. This turns the rate from an open question into an explicit upper bound — and turns the proposal's real content into an experiment. The downstream reaction $n + {}^{10}B \rightarrow {}^7Li + {}^4He$ provides a unique, artifact-resistant triple signature: a 478 keV gamma (the ~94% branch) in coincidence with ${}^4He$ and correlated heat, localized to plasmon hotspots and gate-tunable. Because the threshold field's energy density would rival graphene's cohesive energy, sustained drives stop one to two decades short of ξ_{th}; the experiment is therefore framed as a bound up to the highest survivable drive. The claim is deliberately modest: the mechanism is bounded by theory, and the question is decidable only by measurement.

Authorship and status. *A research-stage proposal, derived from the author's prior work [1] and revised following detailed technical review. Portions were prepared with AI assistance under the author's direction. The author does not hold formal expertise in strong-field QED or nuclear physics, and the document has not undergone independent expert review. The central rate result (§3, Appendix B) is presented as a bound; the experimental program (§9) is the paper's operative contribution. Formal expert review, and the calculations identified in §10, are required before any practical interpretation.*

1. The Problem and the Proposal

The weak process $p + e^- \rightarrow n + \nu_e$ is endothermic by 0.782 MeV when both reactants are at rest. No chemical bond, thermal excitation, or condensed-matter degree of freedom comes within five orders of magnitude of closing that gap — the principal reason that claims of nuclear reactions on hydrogen-loaded surfaces have been met with skepticism for thirty years [2]. One route sidesteps the Coulomb barrier of charged-particle fusion by routing the first nuclear step through neutron formation [7]: an ultra-low-momentum neutron, once created, is absorbed by a neighboring nucleus with no Coulomb obstruction at all. Everything then hinges on a single question — can the capture step itself proceed at a useful rate? Mainstream analyses of the Widom–Larsen proposal concluded that the required mass renormalization is unattainable and the resulting rates negligible [15, 16]; its authors disputed those critiques [17], and the exchange settled nothing experimentally. This paper reproduces the negative verdict by an independent route — closing, in particular, the coherence loophole on which the rebuttals lean — and then reduces what remains to a measurement.

This paper develops the dressed-mass route to that capture step in a graphene/borophane geometry, and then settles its rate. It makes three changes to the Widom–Larsen picture and — in line with the critiques [15, 16], but by an independent argument — establishes the rate as a bound rather than asserting it:

- **A two-dimensional geometry.** Graphene [8] supplies the conduction electrons and the hydride borophane [1, 9] supplies the protons; acoustic graphene plasmons (AGPs) confine terahertz fields far more tightly than any three-dimensional plasmon.
- **A downstream reaction with clean signatures.** The capture product — an ultra-low-momentum neutron — fuses with ^{10}B to give $^{7}Li + {}^{4}He$, releasing charged-particle energy with no hard primary gamma.
- **A regime-robust rate bound.** Section 3 shows that neither raising a single electron's energy nor exploiting plasmon coherence lowers the threshold, in either the multiphoton or the quasistatic (Keldysh) regime, under assumptions stated explicitly in §3.5; the rate is bounded far below detection. The paper therefore leads with the experiment that can decide the question regardless of mechanism.

The logic of what follows is linear. Section 2 fixes the energy gap and introduces the dressed-mass mechanism that could, in principle, open the channel. Section 3 — the technical core — shows by two independent arguments why paying the 0.782 MeV with plasmon quanta fails in every regime, leaving a rate bounded at least tens of thousands of orders of magnitude below detection. Sections 4 and 5 audit the two physical inputs that the bound depends on: the field actually seen by an adsorbed proton, and what graphene plasmons can really deliver. Sections 6–9 then build the downstream reaction and the single coincidence experiment that settles the question of whether the mechanism works, and Section 10 states precisely what is, and is not, being claimed.

2. The Energy Gap and the Dressed-Mass Mechanism

With PDG masses [3], the free-space capture Q-value is

$$Q_{free} = (m_p + m_e - m_n)c^2 = -0.782 \text{ MeV}. \tag{1}$$

A single electron moving at the graphene Fermi velocity carries only about 2.84 eV — smaller than the deficit by a factor of 275,000. The mechanism therefore cannot draw on electron kinetic energy. It relies instead on the intensity-dependent (Volkov) rest mass that a charge acquires in an intense oscillating field [4, 5]:

$$\tilde{m}c^2 = m_e c^2 \sqrt{(1+\xi^2)}, \quad \xi \equiv eA_{rms}/(m_e c). \tag{2}$$

Capture becomes energetically allowed once the dressed mass reaches the neutron–proton splitting, $\tilde{m} \geq m_n - m_p$; equivalently $\xi \geq \xi_{th} = \sqrt{[((m_n - m_p)/m_e)^2 - 1]} \approx 2.32$ (Appendix A). Figure 1 shows the crossing.

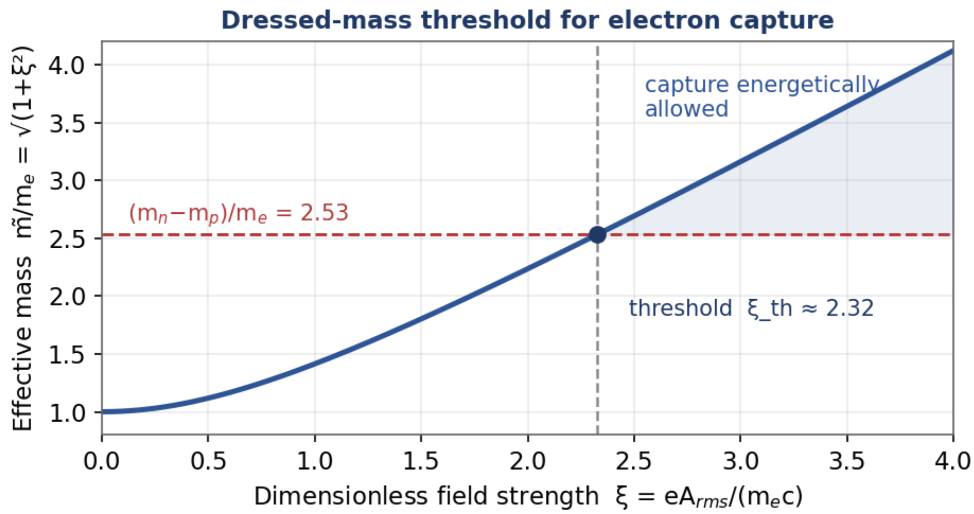

Figure 1. The effective rest mass grows with the dimensionless field strength ξ. Capture becomes energetically allowed once $\tilde{m}/m_e$ exceeds $(m_n - m_p)/m_e = 2.53$, which occurs at $\xi_{th} \approx 2.32$.

At 4 THz this threshold corresponds to $E_{rms} \approx 1.0\times10^{11}$ V·m^{-1} — at the extreme upper edge of reported graphene near-fields [10, 11]. (For calibration, the SLAC E144 experiment operated at $\xi \lesssim 0.4$ [6], well below this regime; it confirmed dressed-electron kinematics but not the conversion of field energy into rest mass for a weak decay.) Whether the field — or the plasmon's coherence — can actually supply the 0.782 MeV is the subject of §3.

Two caveats attach to the threshold itself. First, Eq. (2) is derived for a propagating plane wave — a null field — and E144 calibrates only that case. A plasmon near-field is longitudinal and evanescent, with phase velocity $\omega/q \approx c/120$ here; no theorem transplants the Volkov dispersion to such a mode. The quiver energy is real, but whether it enters the weak-vertex kinematics as rest mass is an assumption of this paper, not a result. Second, the carriers being dressed are Dirac quasiparticles of the graphene band structure; the dressing argument treats the physical electron, of vacuum mass m_e, and uses the band structure only to set the achievable field. Both caveats are inherited by everything that follows: $\xi \geq \xi_{th}$ should be read as conjectural for this geometry — which only sharpens the conclusion of §3, since even the conjecture, granted in full, does not survive the rate analysis.

3. The Rate: Why Coherent Plasmon Coupling Does Not Lower the Threshold

The dressed-mass threshold is real, but reaching it is an energetic statement, not a rate. The energy must be supplied to the weak vertex in quanta of the driving mode, and this section asks whether plasmon coherence can make that affordable. Two independent arguments — one dynamical, one purely kinematic — show that it cannot, and a Keldysh-regime analysis (§3.4) shows that the conclusion is not an artifact of the multiphoton limit.

3.1 The multiphoton barrier

Because energy arrives in quanta of the drive, the process is intrinsically of very high order,

$$N = E_C/\hbar\omega, \quad E_C = 782\ \text{keV}. \tag{3}$$

At a representative AGP energy $\hbar\omega \approx 10$ meV (≈ 2.4 THz), this is $N \approx 7.8\times10^7$. A coherent (classical) field gives the standard Volkov amplitude for absorbing N quanta, the generalized Bessel function $J_N(\xi)$; for $N \gg \xi$ it is hyperexponentially small,

$$|J_N(\xi)|^2 \approx (2\pi N)^{-1}(e\xi/2N)^{2N}.$$

This is the barrier that any rescue must overcome.

3.2 The coherent N-quantum rescue, and why it cancels

The natural rescue treats the AGP as a coherent N-quantum object with an effective vertex,

$$H_{int} = g_{eff}[(a^\dagger)^N\sigma_- + a^N\sigma_+], \quad \Gamma \propto |g_{eff}|^2\langle n^N\rangle \approx |g_{eff}|^2 n^N, \tag{4}$$

from which the required occupation $n^* \gtrsim (\gamma/g_{eff})^{2/N}$ appears to fall toward unity for large N. The error is that g_{eff} is not a free constant. The operator $(a^\dagger)^N$ is generated only at N-th order from the fundamental single-quantum coupling g_0, so

$$g_{eff} \sim g_0^N/\Delta^{N-1}, \quad \Delta \sim \hbar\omega_0 \quad \Rightarrow \quad \Gamma \propto |g_0^N|^2 n^N = (g_0\sqrt{n}/\Delta)^{2N} = \xi^{2N}. \tag{5}$$

In words: coherence does multiply the rate by the bonus n^N, but assembling an N-quantum vertex out of the one-quantum coupling costs exactly g_0^N; and since $\xi \equiv g_0\sqrt{n}/\Delta$, the two factors combine into ξ^{2N} — precisely the multiphoton suppression of §3.1, now viewed from the coherent-state side. The n^N bonus and the g_0^N penalty are the same physics counted once each. Keeping the bonus while treating g_{eff} as order-unity is exactly what produces the false conclusion; restoring the physical coupling returns $|J_N(\xi)|^2$ precisely (Figure 2). A macroscopically occupied plasmon is a coherent state, so this picture does not sit outside the multiphoton result — it reproduces it.

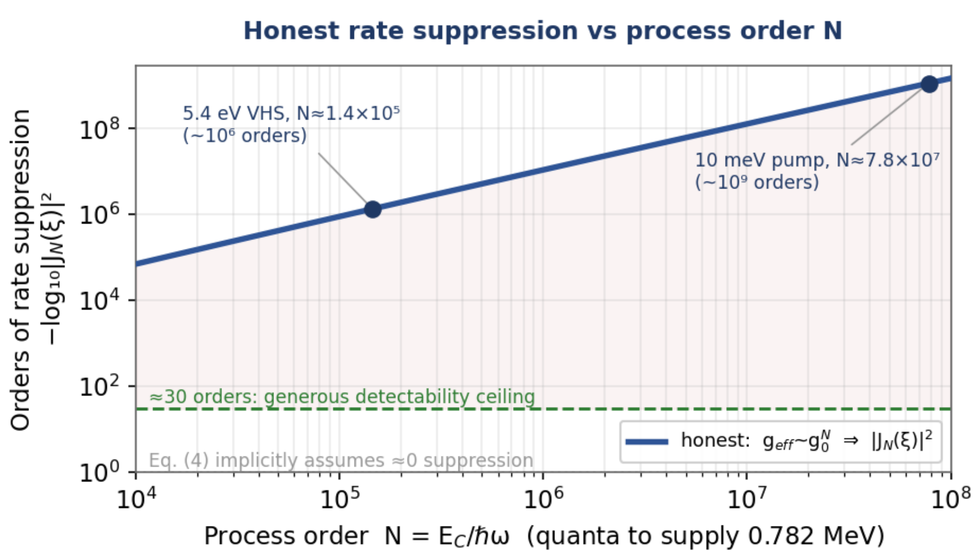

Figure 2. Honest rate suppression $-\log_{10}|J_N(\xi)|^2$ versus process order N, at $\xi = 2.3$. Even the most favorable accessible drive (a 5.4 eV Van Hove resonance, $N \approx 1.4\times10^5$) lies roughly 10^6 orders below any plausible detectability ceiling — and the quasistatic channel (§3.4), though less suppressed ($\approx 6\times10^4$ orders for this drive), is equally decisive; an order-independent coupling, Eq. (4), would place this entire curve at zero.

3.3 The kinematic floor

A second obstruction, free of any dynamical assumption, confirms the result. Because $a^N|n\rangle = \sqrt{[n!/(n-N)!]}|n-N\rangle$ vanishes for $n < N$, the mode must contain at least N quanta before it can supply N. Any expression that returns a required occupation below N is therefore unphysical — yet the coherent-rescue formula does exactly that, giving $n^* \approx 4.3\times10^6$ while $N = 7.8\times10^7$ (Figure 3). Enforcing $n \geq N$ removes the apparent advantage: the minimum occupation is N, and the rate still carries the $|J_N|^2$ suppression.

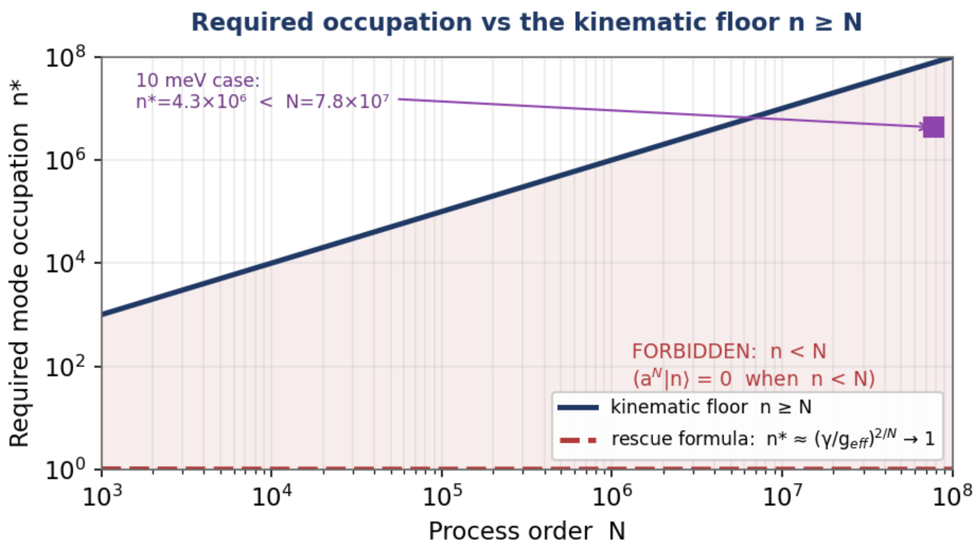

Figure 3. The coherent-rescue formula drives the required occupation n* below the process order N, into the region forbidden by $a^N|n\rangle = 0$. The physical floor is $n \geq N$.

3.4 The quasistatic channel and the Keldysh crossover

The multiphoton suppression of §3.1 is one limit of a more general competition, governed by the Keldysh parameter [18],

$$\gamma_K = \omega\sqrt{(2m_eE_C)}/(eE) = \sqrt{(2E_C/m_ec^2)}/\xi \approx 1.75/\xi, \tag{6}$$

which is independent of drive frequency at fixed ξ and equals 0.75 at threshold. Near and above ξ_{th} the process therefore sits in the quasistatic (tunneling) regime, where the suppression is governed not by $|J_N(\xi)|^2$ but by a Zener-type exponent,

$$\Gamma \propto \exp[-(4/3)\sqrt{(2m_e)}E_C^{3/2}/(eE\hbar)], \tag{7}$$

up to O(1) factors in the exponent (relativistic corrections enter at the same order, since $E_C \approx 1.5\, m_ec^2$). This channel needs no long-lived phase coherence — its formation time is femtoseconds — and it is the least suppressed available. It is still fatal. At the 10 meV threshold field (6×10^{10} V·m^{-1}) the exponent is $\approx -3\times10^7$ in $\log_{10}$; at the 0.20 eV threshold field it is $\approx -2\times10^6$; even at the 5.4 eV Van Hove drive, whose threshold field is 3×10^{13} V·m^{-1}, it is $\approx -6\times10^4$. Table 1 lists both regimes side by side. Quoting the weakest suppression as the operative bound, the most favorable accessible case still falls tens of thousands of orders of magnitude below any detectability ceiling. The bound is regime-robust: no multiphoton-to-tunneling crossover rescues it.

3.5 The bound

Evaluating the honest rate at $\xi \approx 2.3$ across the drives available in a graphene system gives Table 1.

Table 1. Only the single-quantum channel (N = 1, a 0.782 MeV γ-ray) is unsuppressed — and no plasmon or laser exists at that energy. Every coherent THz-to-UV drive is suppressed by 10^6–10^9 orders in the multiphoton regime and by 10^4–10^7 orders in the quasistatic regime (§3.4) — the weaker, and therefore operative, suppression. The final column lists the (unphysical) value that follows from an order-independent coupling.

Coherent drive $\hbar\omega$	$N = E_C/\hbar\omega$	Honest $\log_{10}\lvert J_N(\xi)\rvert^2$	Quasistatic $\log_{10}\Gamma$ (§3.4)	Rescue-formula claim
10 meV (THz plasmon)	7.8×10^7	$\approx -1.2\times10^9$	$\approx -3.4\times10^7$	$n^*\to1$ (4.3×10^6)
0.20 eV (optical phonon)	3.9×10^6	$\approx -5\times10^7$	$\approx -2\times10^6$	$n^*\to1$
5.4 eV (M-point Van Hove singularity)	1.4×10^5	$\approx -1.4\times10^6$	$\approx -6.3\times10^4$	1.2×10^6
0.782 MeV (γ, N = 1)	1	O(1), unsuppressed	O(1)	trivial

Conclusion. None of the available levers crosses zero. Confinement raises g_0 per quantum, but it enters the rate only through ξ, and pushing ξ to order unity is nothing other than the ~10^{11} V·m^{-1} dressed-mass field by another name. The capture rate is bounded at least ~6×10^4 orders of magnitude below detection even in the best accessible case, and by ~3×10^7 for any terahertz drive (§3.4). The bound rests on four explicit assumptions: (i) the weak vertex acts on a single electron; (ii) the drive couples bilinearly to one bosonic mode; (iii) the mode is in a coherent (or less ordered) state; and (iv) decoherence is no slower than measured plasmon lifetimes. Within these assumptions the threshold is simply not lowered; a non-negligible rate would require a genuinely new, strongly correlated many-body channel that violates (i)–(iii) and is not established. We therefore treat the rate as an upper bound (Appendix B) and design the test of §9 so that it does not depend on it.

4. The Field an Adsorbed Proton Sees

The proton does not sit in the bare mode field but in a screened, geometrically reduced local field. Writing $A_{loc} = \eta_{loc}A_0$, the local-field factor η_{loc} is a product of evanescent decay, a bound-state form factor, RPA screening, and a Debye–Waller factor (Appendix C). At $q \approx 10^7$ m^{-1} and an adsorbate height of ~1 Å, the purely geometric reduction is small: $\eta_{loc} \approx 0.999$. Two caveats keep this from settling the matter. First, η_{loc} as computed captures geometric decay only — not the strong-field, many-body screening of a longitudinal field at a charged adsorbate, which could be far larger: static Thomas–Fermi screening at $q \approx 10^7$ m^{-1} in doped graphene already gives $\varepsilon(q) \sim 1 + q_{TF}/q \approx 30$–$100$, i.e. one to two orders of magnitude, before any strong-field treatment. Second, the required ~10^{11} V·m^{-1} meets or exceeds reported dielectric-breakdown fields, and the highest reported amplitudes are sub-nanometer tip-peak values, not coherent fields filling a uniform mode volume. Energetics makes the survival constraint explicit: at 10^{11} V·m^{-1} the field energy density is $\frac{1}{2}\varepsilon_0E^2 \approx 4\times10^{10}$ J·m^{-3}, so even one nanometer of confinement stores ≈ 44 J·m^{-2} — essentially graphene's entire cohesive energy (≈ 45 J·m^{-2} at ~7.4 eV per atom). The lattice does not survive the threshold field; sustainable drives stop one to two decades short. Material survival is therefore a constraint of the proposal, not a loose end (§10). The local-field factor is necessary, but not sufficient, to retire the screening objection.

5. What Graphene Plasmons Deliver

5.1 Dispersion and confinement

Intrinsic graphene plasmons disperse as $\omega \propto \sqrt{q}$ [12]; in the gated geometry relevant here, the acoustic branch is instead nearly linear, $\omega \approx v_A q$ with v_A a few times the Fermi velocity [19]. Either branch gives, at 4 THz, a wavevector $q \approx 10^7\ m^{-1}$ — about 100 times the free-space value. This confinement is exactly what allows two-dimensional plasmons to carry such intense near-fields.

5.2 Field amplitude

Reported amplitudes span several decades [12, 13]: far-field values of $\sim 10^7$–$10^9\ V\cdot m^{-1}$, tip-enhanced near-fields of 10^9–10^{10}, and the highest sub-nanometer tip values of 10^{10}–10^{11}. The dressed-mass threshold sits at the very top of this range — and those peak values are not sustained over a uniform mode volume.

5.3 Coherence

Measured plasmon lifetimes are 0.1–1 ps [10, 11, 14], giving a single-cycle coherence factor $f_{coh} = \omega\tau/(1+\omega\tau)$ of 0.60–0.94 at the representative $\hbar\omega \approx 10$ meV — close to unity over one cycle. But the N-quantum amplitude of §3 requires the drive to stay phase-coherent across all N absorption events, i.e. over $N \approx 8\times10^7$ cycles, or about 30 μs. That is seven to eight orders of magnitude beyond any measured plasmon coherence time. Coherence is thus not a mild correction to the rate; it is part of the central bound. The quasistatic channel of §3.4 evades this coherence requirement — its formation time is far shorter — but is excluded by its own exponent; the two failure modes are complementary.

6. The Downstream Reaction and Its Signatures

Whatever the capture rate, the neutron it produces has a clean fate. An ultra-low-momentum neutron is captured by ^{10}B in the borophane,

$$n + {}^{10}B \rightarrow {}^{7}Li + {}^{4}He \quad (Q \approx 2.79\ \text{MeV}). \tag{8}$$

The thermal $^{10}B(n,\alpha)^{7}Li$ reaction proceeds to the excited $^{7}Li^*$ state in about 94% of events — each emitting a 478 keV gamma — and to the ground state in the remaining ~6%, so the 478 keV line is an abundant diagnostic. (Natural boron contains 19.9% ^{10}B; isotopic enrichment raises the capture fraction accordingly.) Crucially, the reaction releases its energy as charged particles with no hard primary gamma, which is why the 'missing X-ray' objection leveled at metallic-hydride claims does not apply here. The three observable signatures are excess heat, ^{4}He, and the 478 keV gamma.

7. What a Detector Would See

Because the rate is bounded (§3), we do not claim a predicted signal that beats noise. Instead we record the minimum surface activity each detector could see, so that the §9 experiment is framed as either a detection or a quantitative limit. Converting noise floors to an equivalent surface power density gives Table 2.

Table 2. The nuclear-product detectors are six to eleven orders of magnitude more sensitive than calorimetry. (HPGe row: 10^{-3} cps noise in the 478 keV window, 5% absolute efficiency, 94% branch, steady state.) A

calorimetric signal without accompanying ^{4}He and 478 keV gammas would be evidence against the mechanism, not for it.

Detector	Noise floor	Min. detectable surface power
Isothermal calorimeter	~1 mW on 1 cm^2	$\sim 10^{-3}$ W·cm^{-2}
Quadrupole mass-spec (^{4}He)	$\sim 10^{9}$ atoms·cm^{-2}/week	$\approx 7\times10^{-10}$ W·cm^{-2}
HPGe γ-spec (478 keV)	$\sim 10^{-3}$ cps, 5% eff.	$\approx 1\times10^{-14}$ W·cm^{-2}

The quantitative inputs that would set the rate are listed in Table 3, with honest ranges; the surface field dominates, which is what motivates the §9 scan.

Table 3. Parameter ranges. We omit the earlier Monte Carlo 'fraction above threshold,' because it is fixed by the assumed field prior and the bounded rate — making it an input, not evidence.

Input	Plausible range	Status / note
Surface field E_{rms}	10^{7} – 10^{11} V·m^{-1}	Threshold at the extreme upper end; the dominant variable (§9 scan).
η_{loc}	~0.01 – 1.0	Geometric decay ≈ 1; static screening alone may cost 10–10^{2} (§4); strong-field case open.
Plasmon lifetime τ	0.1 – 1 ps	Coherence over N ≈ 8×10^{7} cycles not met (§3, §5.3).
$\|\psi(0)\|^2$ enhancement	10 – 10^{4}	Needs a DFT calculation (§10).
Adsorbate coverage	10^{13} – 10^{15} cm^{-2}	Borophane-class 2D hydrides [9].

8. The Standard Objections, Revisited

- **Screening.** The two-dimensional geometry removes the geometric suppression ($\eta_{loc} \approx 1$); the strong-field, many-body screening of the longitudinal field remains open and should be modeled explicitly.
- **Decoherence.** Real, and — at the N-quantum order required — severe. It is part of the §3 bound, not a separate small effect.
- **Absent X-rays.** Resolved by the boron downstream: no hard primary gamma is expected, and the 478 keV line (the ~94% branch) is itself a diagnostic.

9. A Single Experiment That Could Settle It

One sample; the plasmon drive varied up to the highest survivable amplitude; all three detectors read out in coincidence. No background produces simultaneous excess heat, ^{4}He, and 478 keV gammas in the predicted ratio, so the test is robust against calorimetry artifacts and is mechanism-agnostic: a positive result is a threshold turn-on regardless of which mechanism supplies the energy, while a null result bounds the capture rate at these fields — sharpening, or confirming, the theoretical bound of §3. Because the dressed-mass threshold sits above the survivable window (§4), the experiment cannot cross ξ_{th}; what it can do is bound nuclear activity at every accessible field — and any positive signal there would already imply physics outside the mechanism bounded in §3. The protocol has four parts.

- **Field-amplitude scan.** Vary the drive across 10^{8}–10^{10} V·m^{-1} — the survivable window (§4); the nominal 10^{11} V·m^{-1} threshold lies above it — calibrated with an s-SNOM probe, and look for any super-linear turn-on.
- **Substrate and temperature scan.** Tune the plasmon lifetime through the substrate (free-standing, h-BN, SiO_2) and the temperature (4 K, 77 K, 300 K).
- **Multi-detector coincidence.** Time-correlate calorimetry, ^{4}He mass spectrometry, and a 478 keV HPGe spectrometer; any one signal without the others is evidence against.
- **Null samples.** Borophane-free graphene, graphene-free borophane, and plasmon-off runs must all register no excess.

10. What This Paper Does and Does Not Claim

It does not claim that low-energy electron capture has been observed, nor that the rate is non-negligible — §3 establishes the opposite, as a bound that holds across regimes (§3.4) under the explicit assumptions of §3.5. It does claim three things: that the energy-gap objection alone is incomplete, because the dressed-mass threshold is real physics for null fields, though its transfer to plasmon near-fields remains an assumption (§2); that the coherent N-quantum rescue does not work, for the specific reasons given in §3; and that the downstream chain provides a unique, artifact-resistant signature that makes the question experimentally decidable. Three calculations would close the case, in priority order: (1) a genuine collective/many-body rate estimate — the only route that could change the bound; (2) a DFT calculation of $|\psi(0)|^2$ at the proton site; and (3) a first-principles refinement of the material-survival ceiling estimated in §4. None of them changes the falsifiable prediction that the §9 experiment is built to test. Relative to the existing critiques [15, 16], the contribution here is to close the coherent N-quantum loophole explicitly (§3.2–§3.3), to extend the bound across the Keldysh crossover (§3.4), and to specify a protocol that decides the accessible-field question (§9).

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Appendix A. Threshold and the Dressed Mass

The cycle-averaged Volkov dispersion gives $\tilde{m} = m_e\sqrt{(1+\xi^2)}$. Requiring $\tilde{m}c^2 \geq (m_n - m_p)c^2 = 1.293$ MeV gives $\sqrt{(1+\xi^2)} \geq 2.531$, hence $\xi_{th} = 2.324$.

Appendix B. The N-Quantum Coupling, the Bessel Resummation, and the n ≥ N Floor

The effective N-quantum vertex of Eq. (4) is generated at N-th order in the single-quantum coupling $g_0(a^\dagger\sigma_- + \text{h.c.})$, through N−1 virtual intermediate states, giving $g_{eff} \sim g_0^N/\Delta^{N-1}$ with $\Delta \sim \hbar\omega_0$. Substituting into $\Gamma \propto |g_{eff}|^2\langle n^N\rangle$ yields $\Gamma \propto (g_0\sqrt{n}/\Delta)^{2N} = \xi^{2N}$; resumming all orders gives the generalized Bessel amplitude, $|J_N(\xi)|^2 \approx (2\pi N)^{-1}(e\xi/2N)^{2N}$ for $N \gg \xi$. The n^N coherence bonus is exactly canceled by the g_0^N coupling. Independently, $a^N|n\rangle = 0$ for $n < N$, so the occupation cannot fall below N; the coherent-rescue result $n^* \to 1$ is the artifact of an order-independent coupling. At $\xi = 2.3$, $|J_N|^2 \approx 10^{-1.4\times10^6}$ for $N = 1.4\times10^5$, and $10^{-1.2\times10^9}$ for $N = 7.8\times10^7$. The quasistatic complement to this bound, with its Keldysh crossover at $\gamma_K \approx 0.75$, is given in §3.4.

Appendix C. The Local-Field Factor η_{loc}

For the 2D eigenmode the matrix element factorizes into $F_{height} = e^{-qd}$, a hydrogenic form factor $[1+(qa/2)^2]^{-2}$, RPA impurity screening $[1+4\pi\alpha_{imp}q]^{-1}$, and a Debye–Waller factor $e^{-q^2\langle\delta r^2\rangle/2}$. With $q = 10^7\ \text{m}^{-1}$, d = 1 Å, a = 0.5 Å the product is $\eta_{loc} \approx 0.999$; this captures geometric decay only, as noted in §4.